

CSE400 – Fundamentals of Probability in Computing

Lecture 11: Transformation of Random Variables

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1. Transformation of Random Variables

1.1 Objective

The goal of transformation techniques is to determine the distribution of a new random variable obtained by applying a function to an existing random variable.

Let

$$Y = g(X)$$

where

- X is a random variable with known distribution
- $g(\cdot)$ is a deterministic function
- Y is the transformed random variable

The main problem is:

Given the distribution of X , find the distribution of Y .

1.2 CDF Method for Transformation

The most general technique for transformation is the **CDF method**.

Step 1 — Define the CDF of the new variable

$$F_Y(y) = P(Y \leq y)$$

Since $Y = g(X)$,

$$F_Y(y) = P(g(X) \leq y)$$

Thus, the problem reduces to evaluating this probability in terms of X .

1.3 Case 1 — Monotonic Increasing Transformation

Assume $g(x)$ is **monotonically increasing**, meaning

$$g(x_1) < g(x_2) \quad \text{if} \quad x_1 < x_2$$

Step 1: Express the event

$$F_Y(y) = P(g(X) \leq y)$$

Because g is increasing,

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Thus,

$$F_Y(y) = P(X \leq g^{-1}(y))$$

Step 2: Use the CDF of X

$$F_Y(y) = F_X(g^{-1}(y))$$

Step 3: Obtain the PDF

If X has PDF $f_X(x)$,

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substituting,

$$F_Y(y) = F_X(g^{-1}(y))$$

Using the chain rule,

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

1.4 Case 2 — Monotonic Decreasing Transformation

If $g(x)$ is **monotonically decreasing**, the inequality reverses.

Start with

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

Since g is decreasing,

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus

$$F_Y(y) = P(X \geq g^{-1}(y))$$

Using the CDF definition

$$P(X \geq a) = 1 - F_X(a)$$

Therefore

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

Differentiating,

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

which leads to

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

The absolute value ensures the density remains non-negative.

2. Function of Two Random Variables

Consider transformations involving two random variables.

$$Z = g(X, Y)$$

where X and Y are random variables with joint distribution

$$f_{X,Y}(x, y)$$

The objective is to determine the distribution of the new variable Z .

2.1 Joint Distribution

If X and Y are continuous random variables, they have a joint probability density function $f_{X,Y}(x, y)$.

The probability of a region A is

$$P((X, Y) \in A) = \int \int_A f_{X,Y}(x, y) dx dy$$

3. Illustrative Example

Consider

$$Z = X + Y$$

The goal is to determine the distribution of Z .

3.1 Step 1 — Define the CDF

$$F_Z(z) = P(Z \leq z)$$

Substitute

$$F_Z(z) = P(X + Y \leq z)$$

3.2 Step 2 — Express as a Region

The event $X + Y \leq z$ defines a region in the (x, y) plane.

$$F_Z(z) = \int \int_{x+y \leq z} f_{X,Y}(x, y) dx dy$$

3.3 Step 3 — Evaluate the Integral

The region can be written as

$$y \leq z - x$$

Thus

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y) dy dx$$

3.4 Step 4 — Obtain the PDF

Differentiate the CDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

When X and Y are independent, this leads to the **convolution formula**

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

3.5 Assumption: Independent Random Variables

If X and Y are independent,

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

Substituting this into the previous expression yields the convolution formula.

4. Example Scenario Mentioned in Slides

Let

$$X \sim \text{Exponential}(\lambda)$$

$$Y \sim \text{Exponential}(\lambda)$$

Assume independence and define

$$Z = X + Y$$

Using convolution,

$$f_Z(z) = \int_0^z f_X(x)f_Y(z-x) dx$$

The exponential PDFs are

$$f_X(x) = \lambda e^{-\lambda x}$$

$$f_Y(y) = \lambda e^{-\lambda y}$$

Substituting gives

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Evaluating this integral produces the resulting distribution for the sum.

5. Key Results for Exams

5.1 Transformation of One Random Variable

1. Define the CDF

$$F_Y(y) = P(Y \leq y)$$

2. Substitute the transformation $Y = g(X)$
3. Express the probability in terms of X
4. Differentiate to obtain the PDF

5.2 Monotonic Transformation Formula

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

5.3 Sum of Two Random Variables

For independent X and Y ,

$$Z = X + Y$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

This operation is known as **convolution**.

6. Conceptual Summary

This lecture introduced three fundamental ideas:

- Transformation of random variables using the CDF method
- Handling functions of two random variables using joint distributions
- Computing the distribution of the sum of two variables leading to the convolution formula

These techniques are essential for deriving distributions of new variables constructed from existing random variables.